



Solving complex linear equations

Task A: Introduction to complex linear equations

1) Alex solved this complex linear equation.

Fill in the missing blanks in his work.



$$\frac{7x - 5}{8} = 2$$

$$\frac{2}{1} = \frac{16}{8}$$

$$\frac{7x - 5}{8} = \frac{16}{8}$$

$$7x - 5 = 16$$

$$7x = 21$$

$$x = 3$$

2) Solve:

a) $\frac{2x + 10}{3} = 14$

$$\frac{2x + 10}{3} = \frac{42}{3}$$

$$2x + 10 = 42$$

$$2x = 32$$

$$x = 16$$

b) $\frac{2x}{3} + 10 = 14$

$$\frac{2x}{3} = 4$$

$$\frac{2x}{3} = \frac{12}{3}$$

$$2x = 12$$

$$x = 6$$

Task B: Solving complex linear equations

1) Where is the error in this attempted solution?

Find the error and write a sentence explaining why it is wrong.

$$\begin{aligned} \frac{3x - 5}{2} &= \frac{2x + 1}{10} \\ \frac{3x - 5}{2} \times \frac{1}{5} &= \frac{2x + 1}{10} \\ \frac{3x - 5}{10} &= \frac{2x + 1}{10} \end{aligned}$$

Multiplying the LHS by a fifth and not doing the same to the RHS means our equation has lost equality at this point.

$$3x - 5 = 2x + 1$$

$$x - 5 = 1$$

$$x = 6$$

2) Aisha successfully solved two equations. Fill in the missing blanks in her working.



$$\begin{aligned}\frac{3x-5}{2} &= \frac{2x+1}{10} \\ \frac{3x-5}{2} \times \frac{5}{5} &= \frac{2x+1}{10} \\ \frac{5(3x-5)}{10} &= \frac{2x+1}{10} \\ 5(3x-5) &= 2x+1 \\ 15x-25 &= 2x+1 \\ 13x &= 26 \\ x &= 2\end{aligned}$$

$$\begin{aligned}\frac{5-x}{3} &= \frac{3-x}{5} \\ \frac{5-x}{3} \times \frac{5}{5} &= \frac{3-x}{5} \times \frac{3}{3} \\ \frac{5(5-x)}{15} &= \frac{3(3-x)}{15} \\ 5(5-x) &= 3(3-x) \\ 25-5x &= 9-3x \\ 16 &= 2x \\ x &= 8\end{aligned}$$

Task C: Solving simple rational equations

1) Finish each method to solve this equation in two different ways:

$$\begin{aligned}\frac{54}{e} &= 6 \\ e\left(\frac{54}{e}\right) &= e(6) \\ 54 &= 6e \\ e &= 9\end{aligned}$$

$$\begin{aligned}\frac{54}{e} &= 6 \\ \frac{54}{e} &= 6 \times \frac{e}{e} \\ \frac{54}{e} &= \frac{6e}{e} \\ 54 &= 6e \\ e &= 9\end{aligned}$$

2) Solve:

a) $\frac{84}{x} = 12$

$$\frac{84}{x} = 12 \times \frac{x}{x}$$

$$\frac{84}{x} = \frac{12x}{x}$$

$$84 = 12x$$

$$x = 7$$

b) $11 = \frac{84}{x}$

$$x(11) = x\left(\frac{84}{x}\right)$$

$$11x = 84$$

$$x = \frac{84}{11}$$

c) $-7 = \frac{84}{3x}$

$$-7 \times \frac{3x}{3x} = \frac{84}{3x}$$

$$\frac{-21x}{3x} = \frac{84}{3x}$$

$$-21x = 84$$

$$x = -4$$

